Parallel and Distributed Computing

Lecture 9

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Failure Detection

- Failure detection is the process of identifying when a node or component has failed. It is a critical component of fault tolerance.
- **Techniques for Failure Detection**

1. **Heartbeat Messages**:
   1. Nodes periodically send "heartbeat" messages to indicate they are alive.
   2. If a node fails to send a heartbeat within a specified timeout, it is considered failed.
   3. *Example*: Used in distributed systems like Hadoop.
2. **Timeouts**:
   1. A node waits for a response from another node for a specified period.
   2. If no response is received, the node is considered failed.
   3. *Example*: Used in TCP/IP networks.

Failure Detection

1. **Ping/Echo**:
   1. A node sends a "ping" message to another node and waits for an "echo" response.
   2. If no response is received, the node is considered failed.
   3. *Example*: Used in network monitoring tools.
2. **Gossip Protocols**:
   1. Nodes periodically exchange information about their status with other nodes.
   2. If a node stops gossiping, it is considered failed.
   3. *Example*: Used in distributed databases like Cassandra.

Challenges in Fault Tolerance

1. **Scalability**: Ensuring fault tolerance mechanisms work efficiently in large-scale systems.
2. **Consistency**: Maintaining data consistency across replicated nodes.
3. **Performance**: Minimizing the overhead of fault tolerance mechanisms.
4. **Complexity**: Managing the increased complexity of fault-tolerant systems.

DIstributed commit

Distributed Commit Protocols

Distributed commit protocols ensure that all nodes in a distributed system agree on whether to commit or abort a transaction. The two most common protocols are **Two-Phase Commit (2PC)** and **Three-Phase Commit (3PC)**.

DIstributed commit

Two-Phase Commit (2PC)

- **Description**: A coordinator node ensures that all participant nodes agree to commit or abort a transaction.
- **Phases**:
  - **Prepare Phase**:
    - The coordinator sends a "prepare" message to all participants.
    - Participants perform the transaction and reply with a "yes" or "no" vote.
  - **Commit Phase**:
    - If all participants vote "yes", the coordinator sends a "commit" message.
    - If any participant votes "no", the coordinator sends an "abort" message.

Three-Phase Commit (3PC)

- **Description**: An extension of 2PC that adds a third phase to reduce blocking and improve fault tolerance.
- **Phases**:
  - **Prepare Phase**:
    - Same as in 2PC.
  - **Pre-Commit Phase**:
    - The coordinator sends a "pre-commit" message to all participants.
    - Participants acknowledge the pre-commit message.
  - **Commit Phase**:
    - The coordinator sends a "commit" message to all participants.